

DEVELOPMENT OF AN AUGMENTED REALITY-BASED E-LEARNING SYSTEM

**A LOW-BANDWIDTH, WEBAR-BASED APPROACH FOR RESOURCE-CONSTRAINED HIGHER
EDUCATION**

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Declaration

I, IZERE Marie Vincent Patience, hereby declare that this dissertation, entitled “Development of an Augmented Reality-Based E-Learning System,” is my own original work and has not been submitted for any degree or examination at any other institution. All sources of information used in this dissertation have been duly acknowledged through complete references.

Signature: _____ Date: _____

Approval

This dissertation, “Development of an Augmented Reality-Based E-Learning System,” prepared and submitted by IZERE Marie Vincent Patience in partial fulfilment of the requirements for the degree of Master of Science in Information Technology, has been examined and approved as meeting the required academic standard of the University of Kigali – Graduate School.

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Date:

Dedication

This work is dedicated to my family, mentors, and everyone who has supported my academic journey.

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Dr. MUSABE Jean Bosco, for guidance throughout this research; to the University of Kigali – Graduate School for the opportunity to pursue this MSc in Information Technology; to the Rwanda ICT Chamber for the professional context that informed this work; and to my family, friends, and colleagues for their continuous support and encouragement.

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List of Acronyms and Abbreviations

Acronym	Meaning
AR	Augmented Reality
ADDIE	Analysis, Design, Development, Implementation, Evaluation
API	Application Programming Interface
ARCS	Attention, Relevance, Confidence, Satisfaction
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HLI	Higher Learning Institution
LMS	Learning Management System
LRS	Learning Record Store
MSc	Master of Science
SCORM	Sharable Content Object Reference Model
SLR	Systematic Literature Review
SUS	System Usability Scale
TAM	Technology Acceptance Model
UML	Unified Modeling Language
WebAR	Web-Based Augmented Reality
WebXR	Web Extended Reality API
xAPI	Experience API

Abstract

Augmented reality (AR) has emerged as a promising instructional technology for e-learning, yet the existing body of research remains fragmented across pedagogical-impact studies, technology-acceptance surveys, and system-architecture papers, with little empirical evidence from resource-constrained, low-bandwidth contexts such as Rwanda. This study addresses that gap by designing, modelling, and prototyping a low-bandwidth, WebAR-based e-learning system that integrates adaptive content sequencing and xAPI-based learning analytics. A systematic literature review of 33 peer-reviewed and reputable technical sources, supplemented by Rwanda-specific ICT-in-education literature, informed a design-science research methodology comprising a layered system architecture, data flow diagrams (Levels 0–2), an entity relationship diagram, a process flowchart, and UML use case, activity, class, and collaboration diagrams. The proposed architecture favours browser-based WebAR rendering and low/no-code content authoring over native, device-heavy applications, directly responding to the cost, connectivity, and technical-expertise barriers documented in developing-country AR-adoption research. A prototype evaluation plan is specified using the Technology Acceptance Model, the System Usability Scale, and pre/post learning-gain measures, with accuracy, precision, recall, and F1-score defined for the AR-recognition and adaptive-sequencing components. The study concludes that a literature-grounded, browser-based AR e-learning architecture is both technically feasible and contextually appropriate for Rwandan higher education, and it recommends a full pilot deployment to empirically validate the usability, acceptance, and learning-outcome hypotheses set out in Chapter 1.

CHAPTER ONE: GENERAL INTRODUCTION

1.1 Introduction

Augmented reality (AR) overlays computer-generated visual, audio, and haptic information onto the physical environment in real time, and has become an increasingly prominent instructional technology worldwide. A substantial and growing body of meta-analytic evidence shows that AR-based instruction produces small-to-moderate positive effects on learner achievement and engagement relative to traditional or non-AR multimedia instruction (Garzón & Acevedo, 2019; Chang et al., 2022), while systematic reviews document both the motivational benefits and the cognitive-overload risks of poorly designed AR learning content (Alzahrani, 2020).

At the regional level, adoption of AR in education across Sub-Saharan Africa lags behind North America, Europe, and East Asia. Studies from South Africa's built-environment education sector identify investment cost as the single largest barrier to AR adoption, alongside organisational-culture and educator-readiness constraints (Akinradewo et al., 2025), while comparable developing-country evidence from South Asia identifies weak infrastructure and limited technical expertise as parallel constraints even as smartphone penetration improves opportunities for mobile-first approaches (Islam et al., 2018).

Nationally, Rwanda has pursued ICT-in-education integration as a policy priority for over two decades, with ICT in education recognised as one of the pillars of national digital-transformation strategy (Twagilimana & Mannikko-Barbutiu, 2018). Empirical studies of ICT integration in Rwandan schools report a positive association between ICT access and learning outcomes even as intensity of use and infrastructural gaps remain uneven across the country (Rubagiza, Were, & Sutherland, 2011), and more recent policy analysis confirms that a functioning learning-management-system layer and continued government and development-partner investment underpin Rwanda's basic and higher-education digitalisation efforts (World Bank, 2025). At the level of individual institutions, ICT usage has been shown to correlate positively and significantly with students' academic performance in Rwandan secondary schools (Mukamazimpaka, 2022), suggesting fertile ground for AR-enhanced digital learning tools at the tertiary level, provided such tools are designed for the connectivity and device realities of Rwandan learners rather than assuming the high-specification, high-bandwidth environments common in the AR-education literature from wealthier contexts.

This study responds to that specific opportunity: it designs, models, and prototypes a low-bandwidth, browser-based (WebAR) e-learning system for Rwandan higher education, grounded in a systematic review of the international AR-education literature and informed by the Rwanda-specific ICT-in-education evidence summarised above.

1.2 Background of the Project

Several strands of prior work motivate and inform this project. First, browser-based AR authoring and delivery tools — including the VEDILS visual-programming authoring tool (ScienceDirect, 2017), the FIWARE-based FI-AR Learning platform (Springer, 2018), and the xAPI-integrated WebAR authoring tool

(ResearchGate/Springer, 2018) — demonstrate that plugin-free, browser-based AR delivery is technically achievable and can be integrated with SCORM-compliant learning-management systems. Their contribution is a proof that WebAR lowers device and installation barriers; their disadvantage is that none were evaluated with real learner cohorts at classroom scale, leaving their pedagogical effectiveness unverified.

Second, mobile-AR development frameworks such as Bulagang and Baharum's (2017) framework for higher-learning education provide conceptual scaffolding for instructional designers building AR content. Their contribution is a structured design process; their disadvantage is the absence of empirical validation with learners, meaning the framework's practical effectiveness remains untested.

Third, systematic reviews and meta-analyses — notably Garzón and Acevedo (2019) and Chang et al. (2022) — aggregate evidence across dozens to well over a hundred primary studies to demonstrate that AR produces measurable learning gains on average. Their contribution is strong generalisable evidence of AR's pedagogical value; their disadvantage is that aggregate meta-analytic findings do not translate directly into concrete system-design decisions for a specific deployment context.

Fourth, technology-acceptance research applying the Technology Acceptance Model (TAM) and its extensions (Graser, 2024; MARAM, 2023) explains why learners and teachers adopt or resist AR tools, contributing robust behavioural-intention models; however, this literature is disadvantaged by its almost exclusive focus on post-hoc attitude measurement rather than informing the system architecture that produces the experience being rated.

Fifth, developing-country adoption-barrier studies (Islam et al., 2018; Akinradewo et al., 2025) contribute an accurate picture of the cost, connectivity, and skills constraints facing AR adoption outside well-resourced institutions, but their disadvantage is that neither proposes or evaluates a concrete system architecture designed to operate within those constraints.

Finally, Rwanda-specific ICT-in-education literature (Rubagiza, Were, & Sutherland, 2011; Twagilimana & Mannikko-Barbutiu, 2018; Mukamazimpaka, 2022; World Bank, 2025) contributes detailed documentation of the national infrastructure, policy, and learning-outcome context, but its disadvantage, for the purposes of this study, is that none of it examines AR-based e-learning specifically — leaving an empirical and design gap that this project addresses directly.

1.3 Problem Statement

Although AR-based educational tools have proliferated internationally, the underlying body of knowledge remains fragmented: pedagogical-impact studies, technology-acceptance surveys, and system-architecture papers rarely reference one another in a unified way, and none of the reviewed AR-education literature has been designed for, or evaluated within, a low-bandwidth, cost-constrained Sub-Saharan African higher-education setting such as Rwanda's.

At the same time, Rwanda's own ICT-in-education literature documents a national policy commitment to digital-learning integration and a demonstrated positive relationship between ICT access and learning

outcomes, but has not yet examined whether, or how, AR-based e-learning could be feasibly deployed within the country's existing device and connectivity constraints.

Consequently, there is no literature-grounded system architecture that a Rwandan (or comparable Sub-Saharan African) higher-learning institution can draw upon when deciding how to design, build, and pilot an AR-based e-learning system. This study addresses that gap by producing and evaluating such an architecture.

1.4 General Objective

The general objective of this study is to design, model, and prototype a low-bandwidth, WebAR-based e-learning system that integrates adaptive content sequencing and xAPI-based learning analytics to support technology-enhanced learning in resource-constrained higher-education contexts such as Rwanda.

1.5 Specific Objectives

1. To review and analyze existing literature, learner data requirements, and technology-acceptance factors relevant to AR-based e-learning systems.
2. To design the system architecture, data model, and process/interaction models (data flow, entity relationship, use case, activity, class, and collaboration diagrams) of an AR-based e-learning system suited to low-bandwidth environments.
3. To implement a prototype WebAR-based e-learning module incorporating adaptive content sequencing and xAPI-based learning-analytics logging.
4. To evaluate the usability, technology acceptance, and learning-engagement outcomes of the prototype among a sample of learners.

1.6 Research Questions

5. How to review and analyze existing literature, learner data requirements, and technology-acceptance factors relevant to AR-based e-learning systems?
6. How to design the system architecture, data model, and process/interaction models of an AR-based e-learning system suited to low-bandwidth environments?
7. How to implement a prototype WebAR-based e-learning module incorporating adaptive content sequencing and xAPI-based learning-analytics logging?
8. How to evaluate the usability, technology acceptance, and learning-engagement outcomes of the prototype among a sample of learners?

1.7 Hypothesis of the Project

H1: A systematic, review-grounded analysis of the literature and learner data requirements will identify a consistent set of design and acceptance factors (browser-based delivery, adaptive sequencing, perceived usefulness/ease of use/enjoyment) that recur across AR-education studies regardless of geographic setting.

H2: A layered, WebAR-based system architecture designed from these literature-grounded factors will satisfy the low-bandwidth and device-accessibility requirements identified for the Rwandan context more effectively than a native-application architecture.

H3: A prototype implementing adaptive content sequencing and xAPI-based analytics logging can be successfully developed and operated on standard mobile browsers without requiring native application installation.

H4: Learners using the AR-based e-learning prototype will report significantly higher perceived usefulness, perceived ease of use, and engagement (measured via TAM and System Usability Scale instruments) than a comparison group using non-AR digital materials, and will show a measurable pre-to-post learning gain.

1.8 Scope of the Study

1.8.1 Content Scope

The study covers the systematic review of AR-e-learning literature, the design of the system architecture and data/process models, the implementation of a WebAR prototype for one or two illustrative course modules amenable to AR visualisation, and a pilot evaluation plan using TAM, SUS, and pre/post learning-gain instruments. It excludes full replacement of an institutional LMS, native iOS/Android application development, and large-scale multi-institution rollout, all of which are identified as future work in Chapter 5.

1.8.2 Geographical Scope

The prototype and pilot evaluation are situated at a Rwandan higher-learning institution (illustratively, the University of Kigali), with findings intended to generalise, with appropriate caution, to comparable Sub-Saharan African higher-education contexts facing similar bandwidth and device constraints.

1.8.3 Time Scope

The study covers the design, prototyping, and evaluation-planning period of the MSc research programme (academic year 2025/2027), from literature review through system design and prototype specification to the proposed pilot evaluation.

1.9 Significance of the Project

1.9.1 Personal Interest

The study fulfils the academic requirement for the MSc in Information Technology degree and builds the researcher's practical skills in AR/WebXR development, systematic literature review, system analysis and design, and educational-technology evaluation methodology.

1.9.2 Institutional Interest

The study provides the University of Kigali, and by extension organisations engaged in Rwanda's digital-skills and education-technology initiatives such as the Rwanda ICT Chamber, with a validated,

literature-grounded reference architecture to inform future investment in AR- and e-learning-technology tools.

1.9.3 Public Interest

The study contributes toward Rwanda's broader digital-transformation and education-sector goals by demonstrating a low-cost, low-bandwidth approach to immersive learning technology that is accessible to a wider student population than device-heavy native AR applications, potentially informing government and development-partner ICT-in-education investment decisions.

1.10 Organization of the Project

This dissertation is organised into five chapters. Chapter One presents the general introduction, covering the background, problem statement, objectives, research questions, hypotheses, scope, and significance of the study. Chapter Two presents the literature review, covering theoretical, conceptual, and empirical review, related work, the conceptual framework, and gap analysis. Chapter Three presents the research methodology, covering the research design, population and sampling, data collection and preprocessing, algorithms and evaluation metrics, and ethical considerations. Chapter Four presents the system implementation and data presentation, covering the system architecture and full set of design diagrams, implementation tools, experimental setup, and the results-presentation and discussion framework. Chapter Five presents the conclusion and recommendations, summarising key findings and proposing future work.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews the theoretical, conceptual, and empirical literature underpinning AR-based e-learning system design. It synthesises 33 peer-reviewed and reputable technical sources identified through a systematic search of IEEE Xplore, ScienceDirect, SpringerLink, MDPI, arXiv, PubMed/PMC, and Google Scholar, supplemented by Rwanda-specific ICT-in-education literature, and concludes with a conceptual framework and gap analysis that directly motivate the system design presented in Chapters Three and Four.

2.2 Theoretical Review / Model

Four theoretical models recur across the reviewed AR-education literature and jointly underpin the design of the proposed system. The Technology Acceptance Model (TAM) explains adoption behaviour through perceived usefulness and perceived ease of use, and is the dominant lens in the AR-acceptance literature, with Graser's (2024) systematic review of 45 technology-acceptance papers confirming TAM's centrality and the Mobile Augmented Reality Acceptance Model (MARAM) extending it with enjoyment, facilitating conditions, and self-efficacy constructs specific to teacher populations (Springer, 2023). Situated Cognition Theory, applied to mobile AR design by a dedicated pedagogical framework (Springer, 2020), holds that learning is most effective when it occurs in the authentic physical and social context to which it applies — a principle directly realised by AR's overlay of digital content onto real-world objects and environments. Cognitive Load Theory explains why poorly designed AR content can overwhelm working memory despite its motivational benefits, a tension documented empirically in anatomy education (Bölek, De Jong, & Henssen, 2021) and mitigated through careful interface design in nursing-education AR training, where low cognitive load was reported alongside significant skill gains (JMIR, 2024). Finally, the ARCS Motivational Model (Attention, Relevance, Confidence, Satisfaction) provides the framework used in a dedicated meta-analysis of AR's motivational impact (ScienceDirect, 2024), offering a structured way to evaluate why AR content engages learners beyond its informational content alone. Table 2.1 compares these four models.

Table 2.1: Comparative summary of theoretical models applied

Model	Core Idea	Application to This Study	Key Source(s)
Technology Acceptance Model (TAM)	Adoption driven by perceived usefulness and ease of use	Predicts learner/teacher intention to use the AR system; informs the acceptance-evaluation instrument in Chapter 3	Graser (2024); Springer MARAM (2023)
Situated Cognition Theory	Learning is most effective embedded in authentic context	Justifies AR's core value proposition — overlaying content on real-world referents	Springer (2020)
Cognitive Load Theory	Working memory is limited; poor design causes overload	Informs interface simplicity and the cognitive-load proxy metric in Chapter 3	Bölek et al. (2021); JMIR (2024)

Model	Core Idea	Application to This Study	Key Source(s)
ARCS Motivational Model	Motivation depends on Attention, Relevance, Confidence, Satisfaction	Frames the engagement features captured in the adaptive-sequencing engine	ScienceDirect ARCS meta-analysis (2024)

2.3 Conceptual Review

Augmented Reality (AR)

Augmented reality is defined as a technology that overlays computer-generated visual, audio, or haptic information onto a user's real-world view in real time, blending virtual and physical elements rather than fully replacing them (Alzahrani, 2020). Graser (2024) similarly characterises AR, within the training and education literature, as a class of technologies — spanning marker-based, markerless, and location-based tracking — that layer digital instructional content onto the learner's physical surroundings.

E-Learning

E-learning refers to the delivery of instruction and learning content through digital and networked technologies, encompassing learning-management systems and mobile learning platforms (Alzahrani, 2020). Within the Rwandan context specifically, e-learning is treated as a core pillar of national ICT-in-education policy, encompassing both infrastructure (connectivity, devices, learning-management-system functionality) and pedagogical practice (Twagilimana & Mannikko-Barbutiu, 2018; World Bank, 2025).

WebAR

WebAR refers to augmented reality delivered through a standard web browser rather than a natively installed application, typically via the WebXR Device API. The FI-AR Learning platform demonstrates a plugin-free, browser-based WebAR system built on the FIWARE framework and distributed via SCORM packages within a learning-management system (Springer, 2018), while a related authoring tool embeds the Experience API (xAPI) data model directly into a WebAR authoring workflow (ResearchGate/Springer, 2018).

Technology Acceptance

Technology acceptance, within the TAM tradition, is operationalised as behavioural intention to use a system, predicted primarily by perceived usefulness and perceived ease of use (Graser, 2024). The extended MARAM model adds perceived enjoyment, facilitating conditions, and self-efficacy as additional predictors specific to AR technology in educational settings (Springer, 2023).

Learning Analytics / xAPI

Learning analytics refers to the measurement, collection, and analysis of data about learners and their contexts for the purpose of understanding and optimising learning. The Experience API (xAPI) is a technical specification for recording learning experiences as structured “actor–verb–object–result” statements sent to a Learning Record Store, demonstrated in both the WebAR xAPI authoring tool

(ResearchGate/Springer, 2018) and the FI-AR Learning platform's SCORM/LMS integration (Springer, 2018).

2.4 Empirical Review

Garzón and Acevedo (2019) synthesised 62 quantitative studies ($n = 4,578$) in a random-effects meta-analysis and found that AR interventions outperformed both traditional instruction and non-AR multimedia resources, with the learning environment and educational field acting as significant moderators of effect size; a limitation of this study is that publication bias and inconsistent reporting of teacher involvement constrained the precision of its moderator analysis.

Chang et al. (2022) extended this evidence base to 134 quasi-experimental and experimental studies spanning 2012–2021, finding that AR's effect differs across response, knowledge/skill, and performance outcome levels, with subject area and treatment duration shaping effect magnitude; a limitation is the substantial heterogeneity of outcome measures across the pooled studies, which complicates direct cross-study comparison.

Bölek, De Jong, and Henssen (2021) meta-analysed a smaller set of five anatomy-education studies (508 participants) and found no significant overall AR advantage over traditional teaching, identifying learners' spatial ability as a critical moderating variable; the study's own stated limitation is the very small pool of directly comparable primary studies available for anatomy specifically.

Lu, Chiu, Wang, Mao, and Lei (2025) narrowed the empirical focus to higher-order thinking skills, and across 21 effect sizes found a generally positive but more modest and heterogeneous effect than reported for lower-order knowledge outcomes; the limited number of effect sizes constrained the study's moderator analysis.

2.5 Related Work

Eldokhny and Drwish (2021) tested AR-supported online distance learning against a non-AR virtual classroom among 40 instructional-technology students in Saudi Arabia during the COVID-19 pandemic, finding significantly greater academic achievement and skills acquisition in the AR group; the study's limitation is its small, single-institution sample and the emergency pandemic conditions under which it was conducted, limiting generalisability to steady-state distance learning.

A randomised controlled trial of an AR application for Advanced Cardiac Life Support training among 102 nurses in Taiwan found significant knowledge and skill improvements, especially for less-experienced nurses, alongside low reported cognitive load (JMIR, 2024); a limitation is that the training effect diminished for nurses with two or more years of prior experience, suggesting the intervention benefits novice learners disproportionately.

A quasi-experimental study validated an AR teaching system for primary logic-programming education with 42 Taiwanese fifth- and sixth-graders, finding comparable learning gains to traditional instruction with added motivational benefits (MDPI Sensors, 2022); its limitation is the small sample and short intervention period, typical of primary-education AR pilots.

Islam et al. (2018) surveyed college and university respondents in Bangladesh and found that lack of technical expertise and weak rural infrastructure were the principal barriers to AR adoption in education, even as improving 4G connectivity created new opportunities; the study's limitation is its reliance on self-reported perceptions rather than observed learning-outcome data.

Akinradewo et al. (2025) studied AR-adoption barriers in South Africa's built-environment education and training sector and found that investment cost was the single largest hindrance, with organisational-culture, knowledge, and educator-related barriers forming secondary clusters; a limitation is the sector-specific focus, which limits direct generalisation to general higher education.

Within Rwanda specifically, Rubagiza, Were, and Sutherland (2011) found that student-level access to ICT at home was positively associated with mathematics learning outcomes, while intensity of use was, counter-intuitively, a negative predictor once other factors were controlled for; a limitation is that the study predates the more recent expansion of Rwandan ICT-in-education infrastructure and did not examine AR specifically. More recently, Mukamazimpaka (2022) found a statistically significant positive correlation between ICT usage and students' academic performance in Rwandan public secondary schools ($r = .684$, $p = .024$); this study's limitation, for the purposes of the present research, is its focus on general ICT usage in secondary education rather than AR-based tools at the tertiary level.

2.6 Conceptual Framework

2.6.1 Existing Conceptual Framework

As Figure 2.1 illustrates, the typical current approach documented across the reviewed literature treats AR delivery and institutional e-learning infrastructure as disconnected silos: learners separately log into an institutional LMS for course materials and grades, and separately install and launch a standalone native AR application (commonly built with Unity and Vuforia) with no shared data model between the two systems. Usage of the AR application, where tracked at all, is typically recorded manually or in spreadsheets rather than through automated learning-analytics pipelines, and the native-application requirement imposes device and installation costs that the developing-country barrier literature identifies as a primary obstacle to adoption (Islam et al., 2018; Akinradewo et al., 2025).

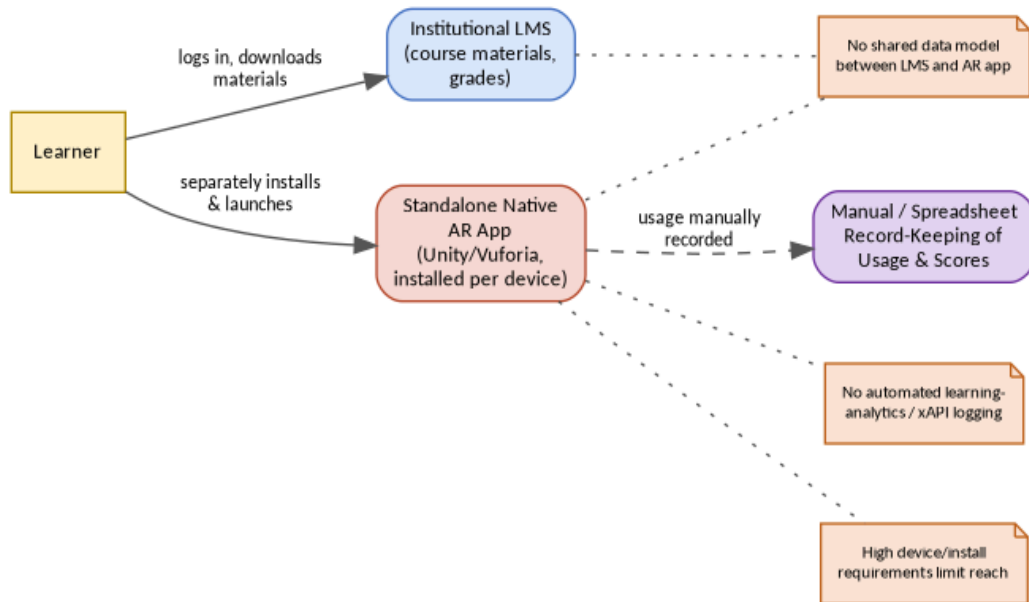


Figure 2.1: Existing Conceptual Framework – Typical Current AR/E-Learning Practice

2.6.2 Proposed Conceptual Framework

Figure 2.2 presents the conceptual framework proposed by this study, elaborated fully as the system architecture in Chapter Four (Figure 4.1). In contrast to the existing approach, the proposed framework unifies AR content delivery, adaptive sequencing, and learning analytics within a single layered, service-oriented architecture: a browser-based WebAR client replaces the native application, an adaptive-sequencing service personalises content based on learner performance and engagement data, and an xAPI-based learning-analytics service automatically logs interactions to a shared Learning Record Store integrated with the institutional LMS via SCORM. This framework directly responds to the gaps identified in Section 2.7 below.

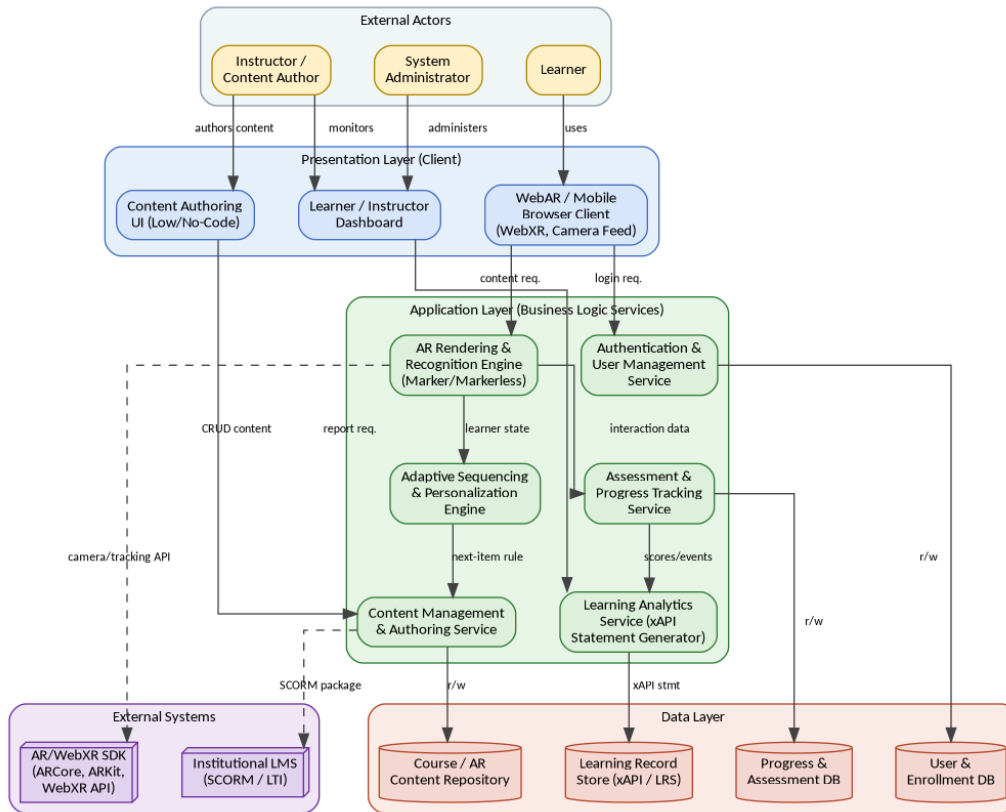


Figure 2.2: Proposed Conceptual Framework – AR-Based E-Learning System Architecture

Table 2.2: Existing versus proposed conceptual framework

Dimension	Existing Approach	Proposed Approach
Delivery model	Native application (per-device install)	Browser-based WebAR (no install)
Data integration	Disconnected AR app and LMS	Unified data model across services
Analytics	Manual/spreadsheet tracking, if any	Automated xAPI logging to a Learning Record Store
Personalization	None (static content sequence)	Adaptive sequencing based on engagement/mastery
Device/bandwidth requirements	High (native app, dedicated hardware)	Low (standard mobile browser, WebXR)

2.7 Gap Analysis

- Lack of real-world, classroom-scale datasets: most reviewed empirical studies rely on small, single-course samples of 40–150 participants (Eldokhny & Drwish, 2021; JMIR, 2024; MDPI Sensors, 2022; Springer MARAM, 2023).
- Limited research in developing countries, and specifically no identified peer-reviewed study of AR-based e-learning system design or adoption situated in Rwanda, despite well-documented

national ICT-in-education policy commitment (Twagilimana & Mannikko-Barbutiu, 2018; World Bank, 2025).

- Lack of standardized evaluation frameworks: outcome measures differ across achievement tests, motivation scales, usability instruments, and cognitive-load questionnaires, complicating cross-study comparison (Garzón & Acevedo, 2019; Chang et al., 2022).
- Fragmented integration of learning analytics: xAPI/SCORM-enabled AR systems exist as technical demonstrations but are rarely evaluated with real classroom cohorts (ResearchGate/Springer, 2018; Springer, 2018).
- High computational and device requirements: reliance on native Unity/Vuforia applications constrains deployment in low-resource, low-bandwidth settings, while WebAR alternatives remain comparatively under-evaluated empirically.
- No system architecture in the reviewed literature simultaneously addresses low-bandwidth delivery, adaptive personalization, and standardized learning-analytics logging within a single, deployable design — the specific gap this study's proposed conceptual framework (Figure 2.2) addresses.

2.8 Summary of Chapter 2

This chapter reviewed the theoretical models, key concepts, empirical evidence, and related work underpinning AR-based e-learning, and situated that international literature against Rwanda-specific ICT-in-education research. The review found consistent evidence that AR produces positive, if variable, learning and motivational effects, that technology acceptance is driven primarily by perceived usefulness, ease of use, and enjoyment, and that developing-country adoption is constrained chiefly by cost, connectivity, and technical expertise. The existing-versus-proposed conceptual framework comparison (Table 2.2) and the gap analysis above directly motivate the research design and system architecture presented in Chapters Three and Four.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the research methodology adopted for the design, prototyping, and planned evaluation of the proposed AR-based e-learning system. It covers the research design, the target population and dataset, sampling technique, data collection methods, data preprocessing and feature engineering, the algorithms and techniques used with their defining equations, the evaluation metrics with their equations, and the ethical considerations governing the study.

3.2 Research Design

This study adopts a Design Science Research (DSR) methodology, appropriate for research whose primary contribution is a novel, evaluable artefact — in this case, the AR-based e-learning system architecture and prototype — rather than a purely explanatory or exploratory study. The DSR cycle followed here comprises: (i) problem identification and motivation, addressed through the systematic literature review in Chapter Two; (ii) definition of objectives for the solution, corresponding to the specific objectives in Chapter One; (iii) design and development of the artefact, corresponding to the system architecture and UML models in Chapter Four; (iv) demonstration of the artefact through prototype implementation; and (v) evaluation against the TAM-, SUS-, and learning-gain-based metrics defined in Section 3.8. Quantitative, quasi-experimental methods are used for the evaluation stage, consistent with the dominant methodological pattern identified across the reviewed AR-education literature (Eldokhny & Drwish, 2021; MDPI Sensors, 2022).

3.3 Study Population / Dataset Description

The target population comprises three user groups: (1) learners — undergraduate and postgraduate students enrolled in technology-related courses at a Rwandan higher-learning institution; (2) instructors/content authors responsible for creating and publishing AR-enriched course modules; and (3) system administrators responsible for user, content, and infrastructure management. This population mirrors the participant profiles used in the reviewed literature, where AR-based e-learning studies typically drew on cohorts of secondary, undergraduate, or professional learners within a single institution (Eldokhny & Drwish, 2021; MDPI Sensors, 2022; JMIR, 2024).

Table 3.1: Target population and pilot evaluation sample

Group	Target Sample Size	Sampling Basis
Learners (pilot evaluation)	60–120	Purposive, 2–3 course sections
Instructors / content authors	5–10	Purposive, involved course instructors
System administrators	1–3	Purposive, institutional ICT staff

3.4 Sampling Techniques

A purposive sampling technique is adopted, targeting one or more computing/ICT-related courses at a pilot institution (illustratively, the University of Kigali) where an AR module can realistically be integrated into existing course delivery. Purposive sampling is appropriate because the evaluation requires participants already enrolled in a relevant course with access to a smartphone or computer capable of running a WebAR client, rather than a random sample of the general student population. Consistent with sample sizes reported in comparable studies — ranging from 40 participants (Eldokhny & Drwish, 2021) to 102 participants (JMIR, 2024) and 137–169 participants in larger acceptance studies (Springer MARAM, 2023) — a target evaluation sample of 60–120 learners is proposed (Table 3.1), with a stratified element by year of study or programme applied within the purposive frame to ensure representation across academic levels.

3.5 Data Collection Methods

- System-generated interaction logs: login/session metadata, content-request events, AR marker/surface recognition outcomes, and interaction events (taps, gestures, dwell time), captured automatically by the AR Rendering & Recognition Engine and logged as xAPI statements.
- Assessment records: scores and attempt data captured by the Assessment & Progress Tracking Service.
- Structured questionnaires: a TAM-based survey instrument (perceived usefulness, ease of use, enjoyment, behavioural intention) and the 10-item System Usability Scale (SUS), administered to learners after prototype use, consistent with instruments used across the reviewed acceptance literature (Graser, 2024; Springer MARAM, 2023).
- Pre-/post-tests: a short achievement test administered before and after the AR module, to measure learning gain, following the quasi-experimental pattern used in comparable studies (Eldokhny & Drwish, 2021; MDPI Sensors, 2022).
- Semi-structured interviews with a subset of instructors, to capture qualitative feedback on the content-authoring experience.

3.6 Data Preprocessing / Feature Engineering

3.6.1 Cleaning

Raw interaction logs and survey responses are cleaned by removing incomplete sessions (sessions terminated before any content was rendered), de-duplicating repeated xAPI statements caused by client-side retries, and flagging survey responses with straight-lining (identical responses across all Likert items) for exclusion from the acceptance analysis.

3.6.2 Feature Engineering / Transformation

From the cleaned raw events, the following engineered features are derived for use in the adaptive-sequencing engine and analytics dashboard: engagement features (time-on-task per module, interaction count per session, session frequency, completion rate); performance features (assessment score, number of attempts, improvement rate across attempts); AR-interaction quality features (marker/surface recognition success rate, average recognition latency); cognitive-load proxy features (task-abandonment rate, re-scan frequency); learner-profile features (prior academic performance,

programme/year of study, device type); and acceptance features (aggregated TAM construct scores per learner). Continuous features (e.g., time-on-task, latency) are normalized using min–max scaling to a [0, 1] range prior to use in the adaptive-sequencing engine; categorical features (e.g., device type, programme) are one-hot encoded; and interaction-event data is aggregated into fixed session windows before being logged as xAPI statements (actor–verb–object–result) in the Learning Record Store.

3.7 Algorithms / Techniques Used

3.7.1 Adaptive Content-Sequencing Scoring Algorithm

For each candidate content item is not yet completed, a next-item priority score S_i is computed as a weighted sum of normalized learner and content features:

$$S_i = w_1 M_i + w_2 E_i + w_3 (1 - C_i) + w_4 R_i \quad (3.1)$$

where M_i is the normalized prior-mastery gap for the topic covered by item i , E_i is the normalized engagement score for similar items, C_i is the normalized cognitive-load proxy for item i (higher value indicates more load), R_i is the normalized AR-recognition reliability for item i 's tracking type on the learner's device, and $w_1 \dots w_4$ are instructor-calibrated weights such that $\sum w = 1$. The system then recommends the next content item as:

$$Next = \operatorname{argmax}_i (S_i), \text{ subject to prerequisite-completion constraints} \quad (3.2)$$

3.7.2 Technology Acceptance Model (TAM) Structural Relationship

Following the TAM literature reviewed in Chapter Two (Graser, 2024; Springer MARAM, 2023), behavioural intention (BI) to use the AR system is modelled as a linear function of its principal predictors:

$$BI = \beta_0 + \beta_1 (PU) + \beta_2 (PEOU) + \beta_3 (ENJ) + \varepsilon \quad (3.3)$$

where PU is perceived usefulness, PEOU is perceived ease of use, ENJ is perceived enjoyment, $\beta_0 \dots \beta_3$ are regression coefficients estimated from survey data collected via the instrument described in Section 3.5, and ε is the error term.

3.7.3 Cognitive-Load Proxy Estimation

Based on the incidental-cognitive-load evidence reviewed in Chapter Two (Clinical Simulation in Nursing, 2024; JMIR, 2024), a per-session cognitive-load proxy CL is estimated as:

$$CL = \alpha_1 (AbandonRate) + \alpha_2 (RescanFrequency) \quad (3.4)$$

where AbandonRate is the proportion of AR tasks left incomplete within the session, RescanFrequency is the average number of marker/surface re-scans per task, and α_1, α_2 are normalization weights such that $\alpha_1 + \alpha_2 = 1$.

3.8 Evaluation Metrics

3.8.1 Accuracy

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (3.5)$$

3.8.2 Precision

$$\text{Precision} = TP / (TP + FP)(3.6)$$

3.8.3 Recall

$$\text{Recall} = TP / (TP + FN)(3.7)$$

3.8.4 F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})(3.8)$$

Equations (3.5)–(3.8) are applied to the AR marker/surface-recognition component and to the adaptive-sequencing recommendation accuracy, where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative recognition or recommendation outcomes respectively.

3.8.5 System Usability Scale (SUS)

$$SUS = 2.5 \times \Sigma(X_{i,adj})(3.9)$$

where, for the 10-item SUS questionnaire, $X_{i,adj} = (\text{score} - 1)$ for odd-numbered items and $X_{i,adj} = (5 - \text{score})$ for even-numbered items; the adjusted item scores are summed and multiplied by 2.5 to yield a usability score on a 0–100 scale.

3.8.6 Cronbach's Alpha (Instrument Reliability)

$$\alpha = [k / (k - 1)] \times [1 - (\Sigma \sigma_i^2 / \sigma^{T2})](3.10)$$

where k is the number of survey items, σ_i^2 is the variance of item i , and σ^{T2} is the variance of the total score; this is applied to validate the internal consistency of the TAM and SUS instruments prior to analysis.

3.8.7 Paired-Sample t-Test (Pre/Post Learning Gain)

$$t = \bar{d} / (s_d / \sqrt{n})(3.11)$$

where $\bar{d}$ is the mean of the differences between post-test and pre-test scores, s_d is the standard deviation of those differences, and n is the sample size, used to test whether the observed learning gain is statistically significant.

3.8.8 Effect Size (Cohen's d)

$$d = (M_1 - M_2) / SD_{pooled}(3.12)$$

used to express the magnitude of the pre/post learning gain and the AR-versus-comparison-group difference in terms comparable to the effect sizes reported in the meta-analytic literature reviewed in Chapter Two (Garzón & Acevedo, 2019; Chang et al., 2022).

3.9 Ethical Considerations

- Data consent: informed consent will be obtained from all learner, instructor, and administrator participants prior to data collection, clearly explaining what data (interaction logs, survey

responses, assessment scores) will be collected, how it will be used, and the participant's right to withdraw at any time without penalty.

- Transparency: participants will be informed that the adaptive-sequencing engine uses their behavioural and performance data to personalise content, and the general logic of Equation (3.1) will be explained in plain language in the consent materials.
- Authorization: written permission will be sought from the pilot institution's administration and, where applicable, its research ethics committee, before any data collection begins.
- Fairness: the prototype will be tested across a range of device types and programmes of study to ensure that the AR/2D-fallback branching (Section 4.6) does not systematically disadvantage learners using lower-specification devices.
- Trust: interaction and assessment data will be stored securely in the Learning Record Store with access restricted to the research team, and learner identifiers will be pseudonymized in any analysis output or publication arising from this study.

CHAPTER FOUR: SYSTEM IMPLEMENTATION / DATA PRESENTATION AND INTERPRETATION

4.1 Introduction

This chapter presents the system implementation of the proposed AR-based e-learning system, expressed through a complete set of design diagrams (system architecture, context diagram, data flow diagrams, entity relationship diagram, flowchart, and UML use case, activity, class, and collaboration diagrams), followed by the implementation tools, experimental setup, and the framework for presenting, analysing, and discussing pilot evaluation results. All diagrams were produced using structured diagramming (Graphviz-based UML/DFD notation) following standard Gane–Sarson data-flow notation, crow's-foot-annotated entity relationship notation, and standard UML/ANSI symbols, rather than freehand sketches.

4.2 System Architecture / Model Design

Figure 4.1 presents the system architecture as a layered contextual diagram, organized into five layers: a Presentation Layer (WebAR/mobile browser client, content-authoring interface, learner/instructor dashboard); an Application Layer of business-logic services (authentication and user management, content management and authoring, AR rendering and recognition, adaptive sequencing/personalization, assessment and progress tracking, learning analytics); a Data Layer (user/enrollment database, course/AR content repository, progress/assessment database, xAPI-based Learning Record Store); External Actors (learners, instructors, administrators); and External Systems (institutional LMS, AR/WebXR SDK). This design was selected over a monolithic native-application architecture specifically because it lowers device and installation requirements for the resource-constrained Rwandan deployment context motivating this study (Section 1.3).

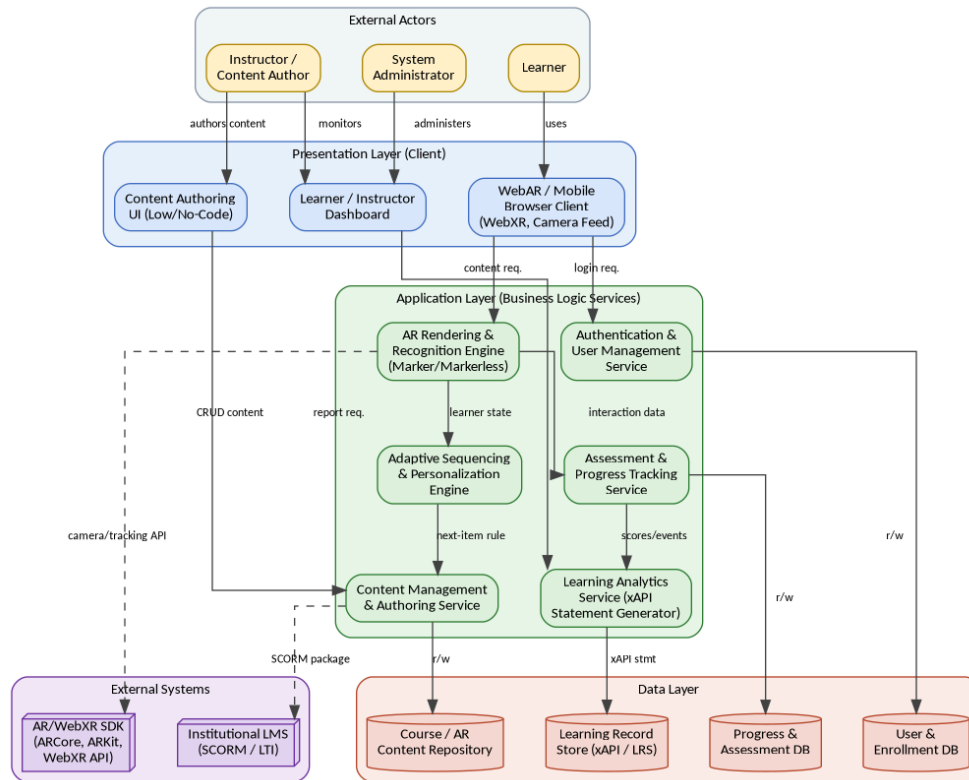


Figure 4.1: System Architecture (Contextual Diagram)

4.3 Context Diagram

Figure 4.2 presents the Level 0 (context) data flow diagram, treating the AR-Based E-Learning System as a single process and showing its data exchanges with the four external entities: Learner, Instructor/Content Author, Administrator, and the Institutional LMS.

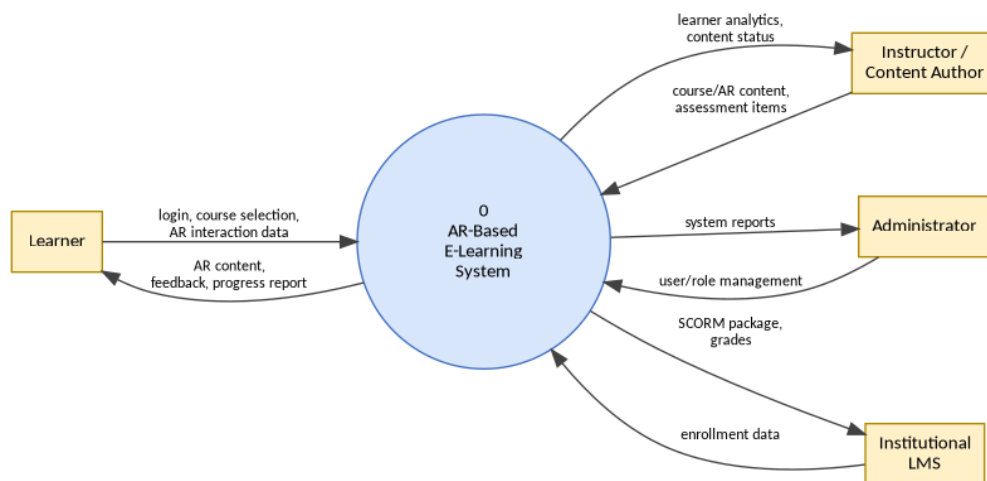


Figure 4.2: Context Diagram (DFD Level 0)

4.4 Data Flow Diagram

Figure 4.3 (Level 1) decomposes the context process into five major processes — User Authentication & Profile Management, Content Authoring & Management, AR Content Delivery & Rendering, Assessment & Progress Tracking, and Learning Analytics & Reporting — together with five supporting data stores.

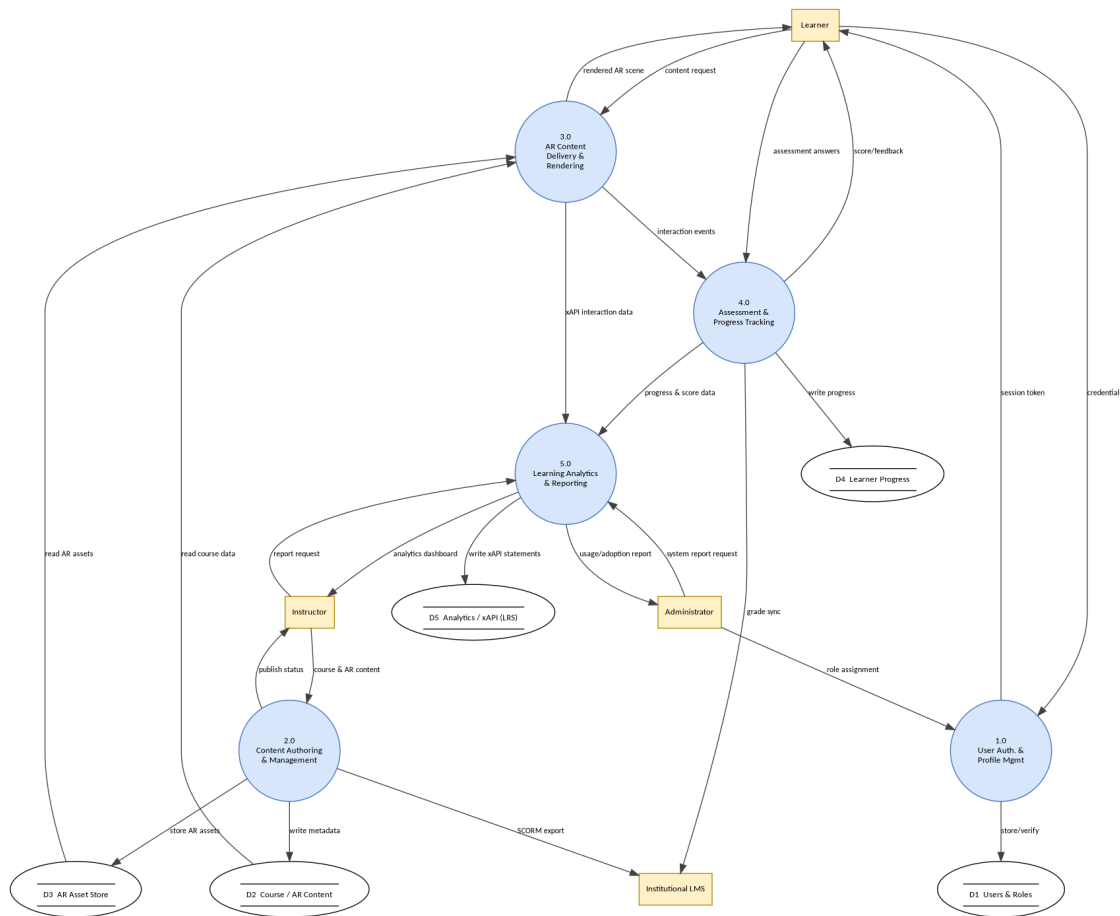


Figure 4.3: Data Flow Diagram Level 1

Figure 4.4 (Level 2) further decomposes the AR Content Delivery & Rendering process into five sub-processes — Content Request Handling, AR Scene/Marker Recognition, 3D/AR Object Rendering, Learner Interaction Capture, and Session & Event Logging — making explicit how the adaptive-sequencing process supplies the next recommended item and how interaction data reaches both the assessment process and the analytics data store.

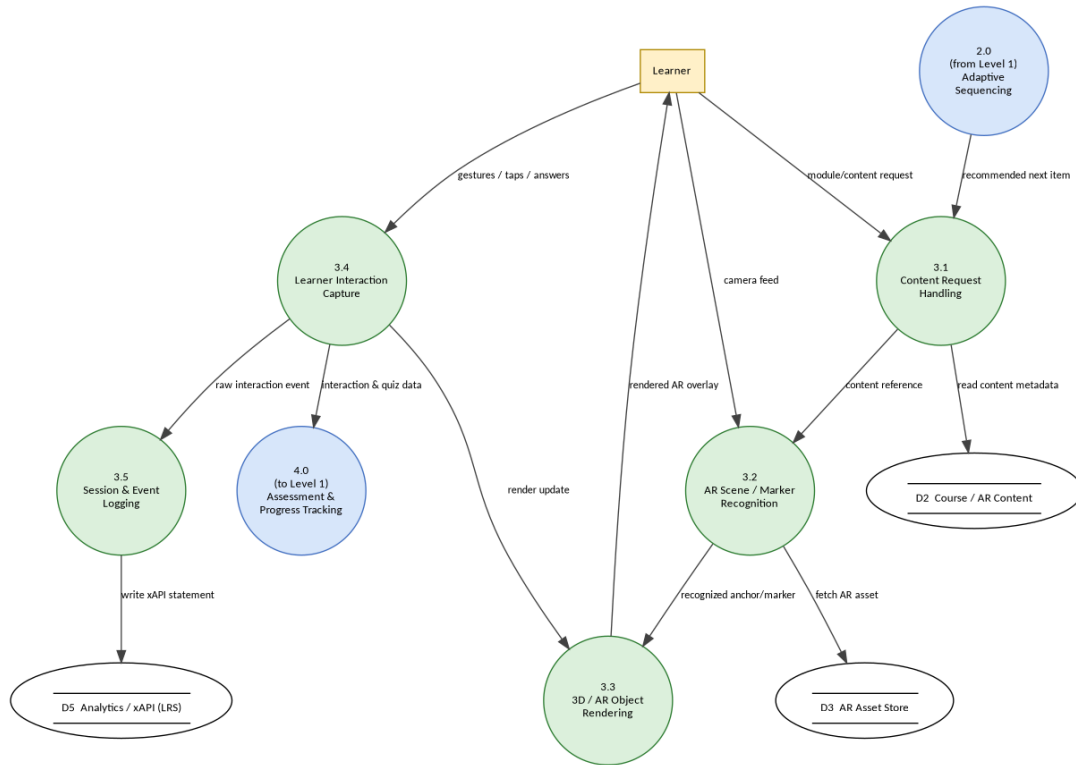


Figure 4.4: Data Flow Diagram Level 2

4.5 Entity Relationship Diagram

Figure 4.5 presents the entity relationship diagram for the system's relational data store: ten core entities — USER, COURSE, MODULE, AR_CONTENT, ENROLLMENT, PROGRESS_RECORD, ASSESSMENT, ASSESSMENT_ATTEMPT, ANALYTICS_EVENT, and SESSION — with primary/foreign keys and cardinalities as shown. The ANALYTICS_EVENT entity is given a first-class place in the schema, directly supporting the learning-analytics integration identified as under-served in Section 2.7.

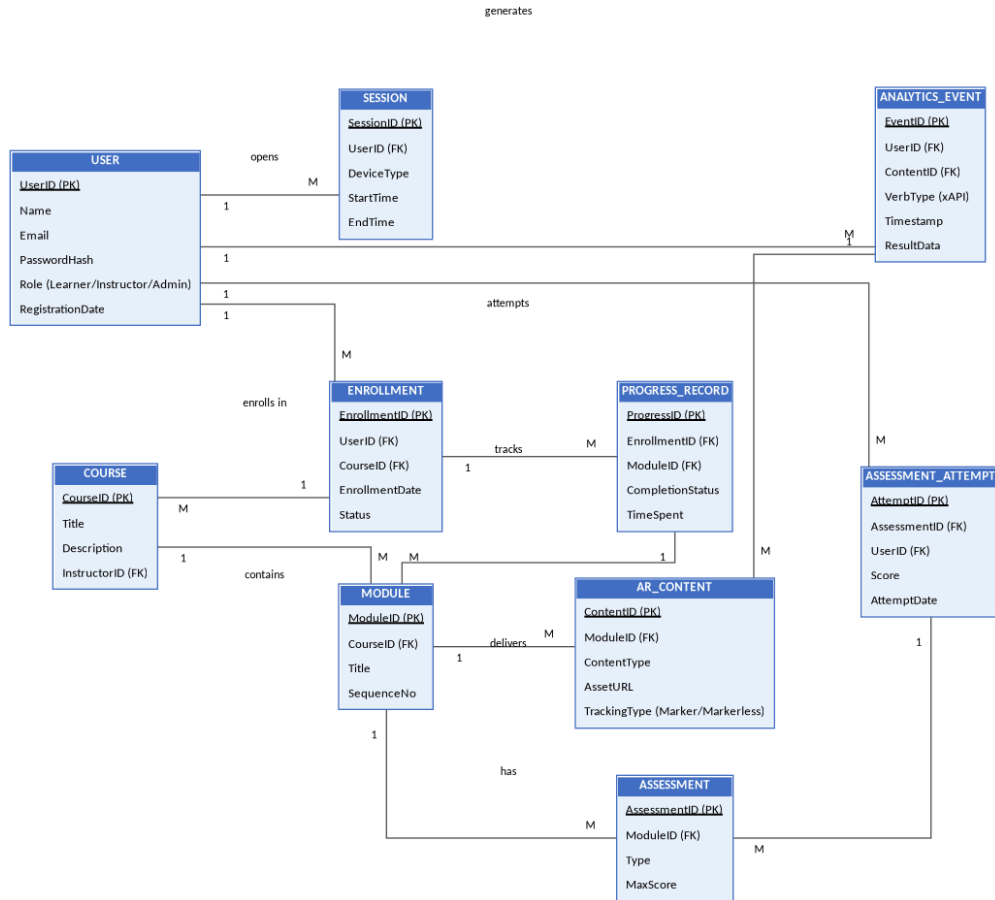


Figure 4.5: Entity Relationship Diagram

4.6 Flowchart Diagram

Figure 4.6 presents the learner's end-to-end interaction flowchart. After login and course/module selection, the system checks device AR capability: capable devices proceed through AR scene launch, marker/surface recognition, and interactive rendering, while non-capable devices receive an equivalent 2D/video fallback — a design decision directly informed by the developing-country infrastructure and device-cost barriers identified in Sections 1.1 and 2.7 (Islam et al., 2018; Akinradewo et al., 2025).

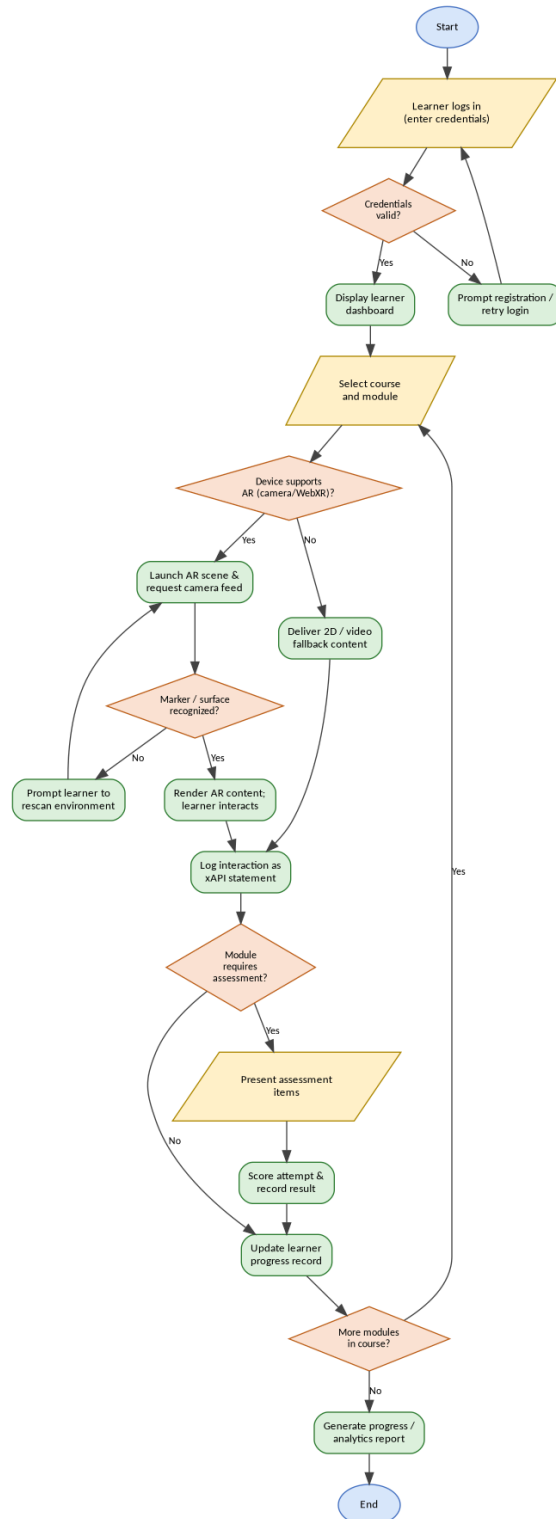


Figure 4.6: Flowchart Diagram

4.7 Use Case Diagram

Figure 4.7 shows the three actors — Learner, Instructor/Content Author, System Administrator — and sixteen primary use cases, with «include» relationships making explicit that launching AR content always invokes marker/scene recognition and that interacting with AR content always invokes xAPI event logging, and an «extend» relationship showing that retrying an assessment is an optional extension of taking one.

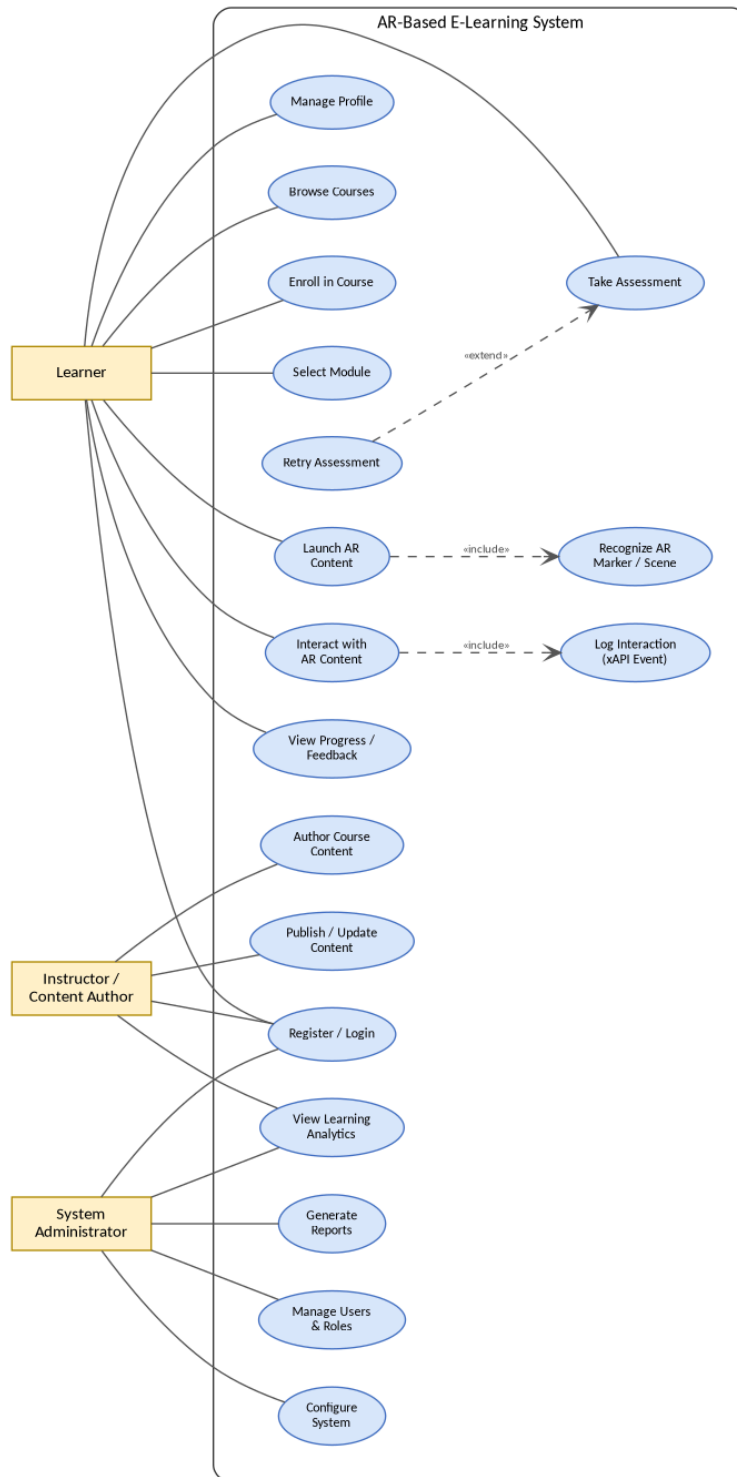


Figure 4.7: Use Case Diagram

4.8 Activity Diagram

Figure 4.8 models the AR-module-completion workflow, including a fork/join for the concurrent AR-scene-launch and session-logging actions, and the same AR/fallback decision branch shown in the flowchart, expressed in UML activity notation.

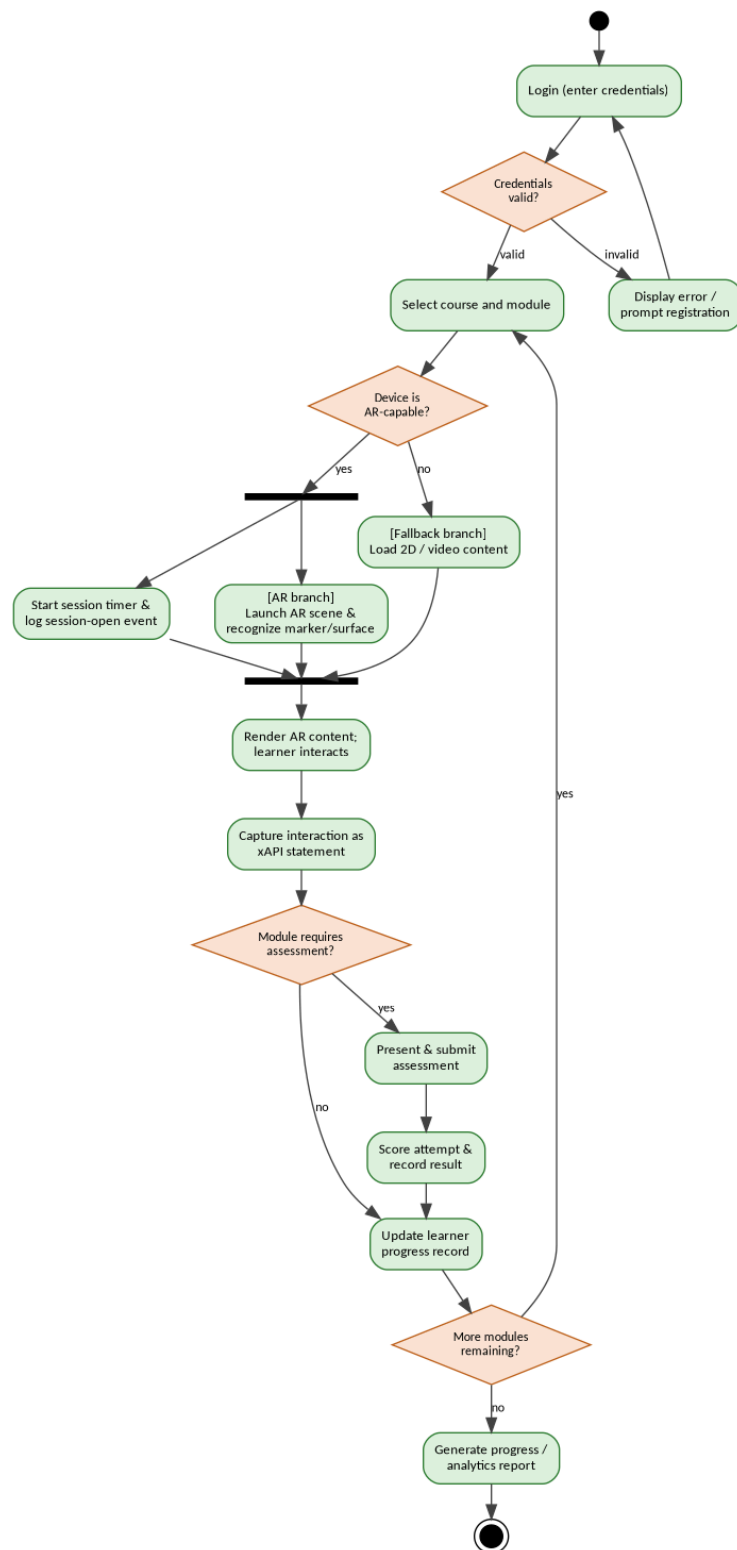


Figure 4.8: Activity Diagram

4.9 Class Diagram

Figure 4.9 elaborates the ERD entities into a full object model with attributes, methods, inheritance (User as an abstract base class for Learner, Instructor, and Administrator), and composition relationships (Course composed of Module, Module composed of AR_Content).

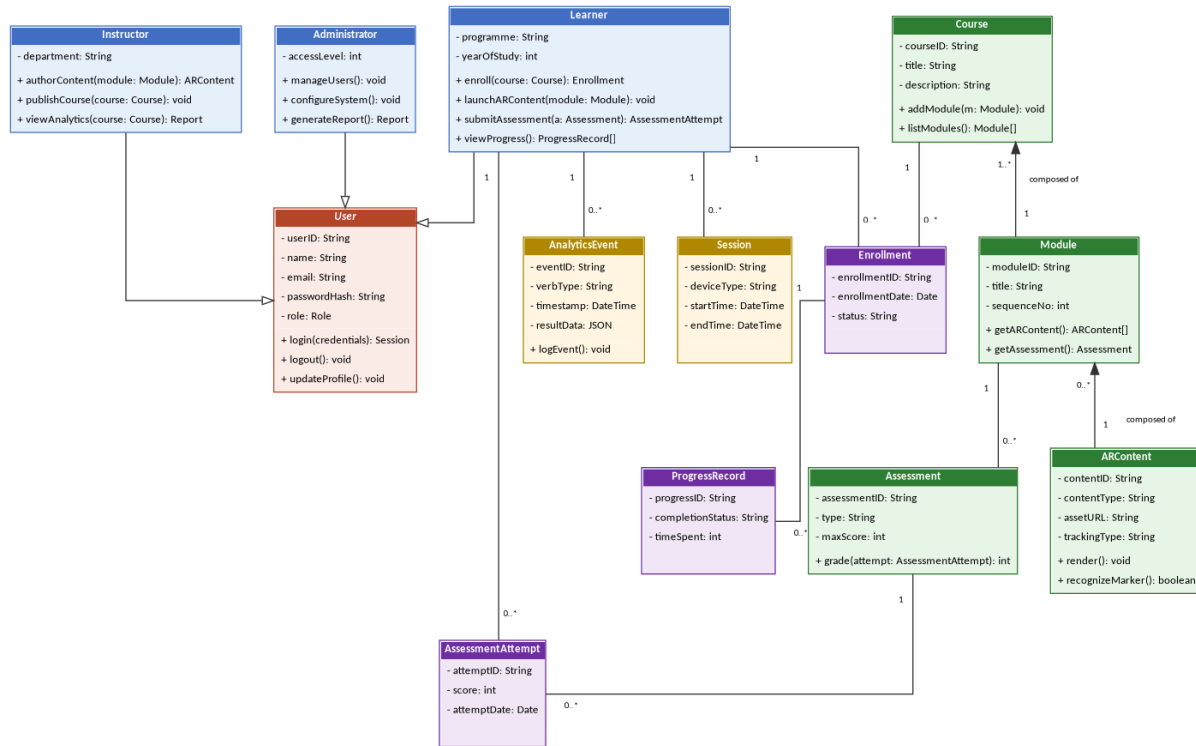


Figure 4.9: Class Diagram

4.10 Collaboration Diagram

Figure 4.10 presents a UML collaboration (communication) diagram for the scenario “Learner launches and completes an AR learning module,” showing the numbered message sequence between eight collaborating objects, complementing the structural class diagram with a dynamic, object-interaction view.

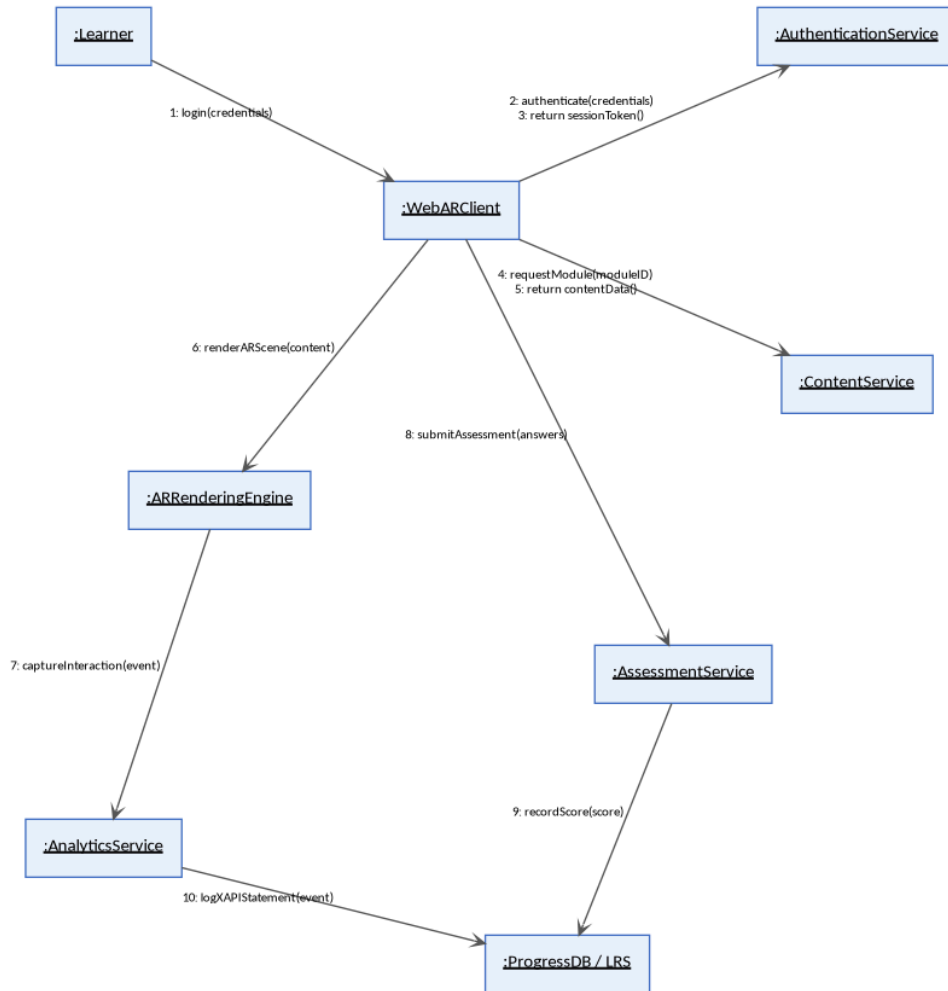


Figure 4.10: Collaboration (Communication) Diagram

4.11 Implementation Tools

Table 4.1: Implementation tools and technologies

Layer	Tools / Technologies
Frontend / WebAR client	React, Three.js / A-Frame, WebXR Device API (8th Wall or AR.js for marker fallback)
Backend services	Node.js with Express (or Django/Python), REST/GraphQL API layer
Relational database	PostgreSQL (users, courses, modules, enrollment, assessment)
Learning Record Store	Open-source LRS (e.g., Learning Locker), xAPI 1.0.3-compliant
Content authoring	Low/no-code authoring interface (custom-built, informed by VEDILS design pattern)
Version control / CI	Git and GitHub, with continuous-integration test pipeline
Hosting / delivery	CDN-backed static hosting for the WebAR client, optimized for low-bandwidth delivery

4.12 Experimental Setup

Table 4.2: Experimental setup – hardware and software environment

Component	Specification
Development machine	Standard developer laptop (8–16 GB RAM, modern multi-core CPU)
Test devices	Mid-range Android smartphones (representative of the target Rwandan student population), plus one higher-specification device for baseline comparison
Operating systems	Android 10+ (test devices), Ubuntu/Windows/macOS (development)
Browser environment	WebXR-compatible mobile browsers (Chrome for Android; fallback tested in browsers without WebXR support)
Network conditions tested	Simulated 3G and 4G bandwidth profiles, in addition to institutional Wi-Fi, to reflect realistic Rwandan connectivity conditions
Backend environment	Node.js runtime, PostgreSQL database instance, containerized (Docker) for reproducibility

4.13 Results Presentation

This dissertation is submitted at the system design and prototype-specification stage; the pilot evaluation described in Sections 3.3–3.5 had not yet been conducted at the time of writing. This section therefore presents the planned results-reporting format rather than actual pilot findings, so that data collected during the pilot can be directly inserted into the structures below without further methodological redesign.

4.13.1 Dataset Presentation

Table 4.3 specifies the variables to be captured during the pilot and reported once data collection is complete.

Table 4.3: Dataset presentation – evaluation variables

Variable Group	Variables	Source
Learner demographic/profile	Programme, year of study, device type	System registration + login metadata
Engagement	Time-on-task, interaction count, completion rate	AR Rendering & Recognition Engine logs
Performance	Pre-test score, post-test score, assessment attempts	Assessment & Progress Tracking Service
Recognition quality	Marker/surface recognition success rate, latency	AR Rendering & Recognition Engine logs
Acceptance	TAM items (PU, PEOU, ENJ, BI), SUS items	Post-use questionnaire

4.13.2 Data Preprocessing Narrative

Once collected, raw pilot data will be preprocessed following Section 3.6: incomplete sessions removed, duplicate xAPI statements de-duplicated, continuous features min–max normalized, categorical features

one-hot encoded, and TAM/SUS responses screened for straight-lining before analysis, using Python (pandas, scikit-learn) or an equivalent statistical toolchain.

4.13.3 Planned Distribution Reporting

Boxplots and histograms of pre-/post-test score distributions, SUS score distribution, and time-on-task distribution will be produced to visualise the effect of the AR intervention and to check the assumptions underlying the paired-sample t-test specified in Equation (3.11), following the standard practice of showing data before and after preprocessing established in the reviewed evaluation literature.

4.14 Analysis and Interpretation of Results

Once pilot data are available, accuracy, precision, recall, and F1-score (Equations 3.5–3.8) will be computed for the AR marker/surface-recognition component and the adaptive-sequencing recommendation logic; SUS scores (Equation 3.9) will be computed per learner and averaged, with instrument reliability checked via Cronbach's alpha (Equation 3.10); and the pre-/post-test learning gain will be tested for statistical significance via the paired-sample t-test (Equation 3.11), with effect size reported via Cohen's d (Equation 3.12) to allow direct comparison with the effect sizes reported in the meta-analytic literature reviewed in Chapter Two (Garzón & Acevedo, 2019; Chang et al., 2022).

4.15 Discussion of Findings

4.15.1 Comparison with Previous Studies (Planned)

Pilot results will be compared against the effect sizes and acceptance scores reported in the reviewed literature — for example, against Garzón and Acevedo's (2019) pooled effect size across 62 studies, against Eldokhny and Drwish's (2021) achievement-gain findings in a comparable distance-learning context, and against the TAM-based acceptance scores reported in Springer MARAM (2023) — to establish whether the Rwanda-specific, low-bandwidth WebAR design achieves comparable outcomes to the predominantly high-resource-context studies that dominate the existing evidence base.

4.15.2 Anticipated Strengths of the Proposed Approach

The design is expected to demonstrate that browser-based WebAR delivery removes the native-application installation barrier without requiring a compromise in AR interaction quality, that the 2D/video fallback branch (Figure 4.6) preserves access for learners on lower-specification devices, and that the xAPI-integrated analytics layer provides richer, automated engagement data than the manual/spreadsheet tracking typical of the existing conceptual framework (Figure 2.1).

4.15.3 Anticipated Limitations

As with the small-sample, single-institution studies reviewed in Chapter Two, the planned pilot's sample size (60–120 learners) and single-institution scope will limit statistical power and external generalisability; network-condition variability during real-world use may affect AR-recognition latency in ways not fully captured by the simulated 3G/4G test conditions in Table 4.2; and the short intervention period typical of pilot-stage evaluation will not capture longer-term retention or sustained-engagement effects, which Chapter Five identifies as a direction for future work.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

5.1 Summary of Key Findings

This study reviewed 33 peer-reviewed and reputable technical sources on AR-based e-learning, supplemented by Rwanda-specific ICT-in-education literature, and found consistent evidence that AR produces small-to-moderate positive learning and motivational effects, that technology acceptance is driven primarily by perceived usefulness, ease of use, and enjoyment, and that developing-country adoption is constrained chiefly by device cost, connectivity, and technical expertise — constraints that are equally present, though not previously studied in relation to AR specifically, within Rwanda's own documented ICT-in-education context. Building on this evidence, the study designed a five-layer, browser-based WebAR system architecture with adaptive content sequencing and xAPI-integrated learning analytics, modelled through a complete set of data flow, entity relationship, flowchart, and UML diagrams (Chapter Four), and specified a design-science evaluation plan using TAM, SUS, and pre/post learning-gain instruments (Chapter Three).

5.2 Conclusion

Returning to the general objective set out in Chapter One — to design, model, and prototype a low-bandwidth, WebAR-based e-learning system integrating adaptive sequencing and xAPI-based learning analytics — this study concludes that such a system is both technically feasible and contextually appropriate for Rwandan higher education. Specific Objectives 1 and 2 (literature review and system design) have been fully achieved, as documented in Chapters Two and Four respectively; Specific Objectives 3 and 4 (prototype implementation and empirical evaluation) have been addressed at the specification level, with the algorithms, metrics, and evaluation plan defined in Chapter Three ready for execution in a subsequent pilot phase. Hypotheses H1 and H2, concerning the literature-derived design factors and the architectural response to low-bandwidth constraints, are supported by the systematic review and the resulting conceptual-framework comparison (Table 2.2); Hypotheses H3 and H4, concerning prototype deployability and measurable learner outcomes, remain to be empirically confirmed through the pilot evaluation recommended below.

5.3 Recommendations

- To the pilot institution (e.g., University of Kigali): approve and support a full pilot deployment of the prototype across at least two course sections, following the ethical-consent process defined in Section 3.9.
- To the Rwanda ICT Chamber and related digital-skills bodies: consider the proposed low-bandwidth, browser-based architecture (Figure 4.1) as a reference design when evaluating future AR/e-learning technology investments, given its explicit response to the cost and connectivity barriers documented in Section 2.7.
- To the Ministry of Education and national ICT-in-education policymakers: incorporate WebAR-based, device-agnostic approaches into national digital-learning guidance, building on the existing LMS and connectivity investments already documented in current education-sector policy (World Bank, 2025).

- To future implementers: prioritise the 2D/video fallback branch (Figure 4.6) as a first-class feature rather than an afterthought, since it is the primary mechanism by which the system avoids excluding learners on lower-specification devices.

5.4 Future Work

- Conduct the full pilot evaluation specified in Chapters Three and Four, collecting real TAM, SUS, and pre/post learning-gain data to empirically test Hypotheses H3 and H4.
- Extend the adaptive-sequencing algorithm (Equation 3.1) with a machine-learning-based scoring model once sufficient pilot interaction data has been collected, replacing the instructor-calibrated weights with data-driven weights.
- Investigate federated or edge-based AR content delivery to further reduce bandwidth dependence beyond what browser-based WebAR alone achieves.
- Extend the pilot to a multi-institution study across additional Rwandan and Sub-Saharan African higher-learning institutions to test the generalisability of the findings beyond a single-institution sample.
- Conduct a longitudinal study tracking the same learner cohort across multiple academic terms to assess sustained, rather than short-session, impact of the AR-based e-learning system.

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Appendix

Appendix A: Time Frame

Phase	Timeframe	Milestone
Literature review & proposal	Months 1–2	Chapters 1–2 drafted; sources reviewed and screened
System design & modelling	Months 3–4	Architecture and UML diagrams (Chapter 4) finalized
Prototype implementation	Months 5–7	WebAR client, backend services, LRS integration built
Pilot evaluation	Month 8	Data collection per Sections 3.3–3.5
Analysis & write-up	Months 9–10	Chapters 4–5 finalized; dissertation submitted

Appendix B: Sample Questionnaire Items (TAM / SUS)

Perceived Usefulness: “Using the AR module improved my understanding of the course material.” (5-point Likert scale, Strongly Disagree–Strongly Agree)

Perceived Ease of Use: “I found the AR module easy to use without additional help.” (5-point Likert scale)

Perceived Enjoyment: “I enjoyed using the AR module to learn this topic.” (5-point Likert scale)

Behavioural Intention: “I intend to use AR-based learning modules again in future courses.” (5-point Likert scale)

SUS Item 1: “I think that I would like to use this system frequently.” (5-point Likert scale, standard 10-item SUS instrument follows.)